

NLP Introduction & Text Processing

Assignment

Question 1: What is Computational Linguistics and how does it relate to NLP?

Computational Linguistics is an interdisciplinary field that combines linguistics, computer science, and artificial intelligence to study and develop systems that can understand, analyze, and generate human language. It focuses on creating computational models of language by applying linguistic rules and theories to language processing tasks. Natural Language Processing (NLP) is a practical application of Computational Linguistics that enables computers to interact with humans using natural language. While Computational Linguistics provides the theoretical foundation for understanding language structure, grammar, syntax, and semantics, NLP uses these concepts to build real-world applications such as machine translation, chatbots, speech recognition, sentiment analysis, and text summarization. Therefore, Computational Linguistics and NLP are closely related, with Computational Linguistics serving as the scientific basis and NLP serving as the implementation of language-processing technologies.

Question 2: Briefly describe the historical evolution of Natural Language Processing.

The evolution of Natural Language Processing (NLP) began in the 1950s when researchers started exploring ways for computers to understand and process human language. One of the earliest developments was rule-based machine translation, where linguistic rules and dictionaries were used to translate text between languages. During the 1960s and 1970s, NLP systems relied heavily on handcrafted grammar rules and linguistic knowledge, but these systems were limited because they could not handle the complexity and ambiguity of natural language. In the 1980s and 1990s, the focus shifted toward statistical methods, where computers learned language patterns from large datasets rather than relying solely on predefined rules. This period saw the introduction of probabilistic models, hidden Markov models, and n-gram techniques. In the 2000s, machine learning became widely used in NLP, improving tasks such as text classification, speech recognition, and information retrieval. The 2010s brought deep learning techniques, including neural networks, word embeddings, and recurrent neural networks, which significantly improved language understanding. More recently, transformer-based models such as BERT and GPT have revolutionized NLP by achieving high accuracy in tasks like translation, summarization, question answering, and conversational AI, making human-computer interaction more natural and effective.

Question 3: List and explain three major use cases of NLP in today's tech industry.

Natural Language Processing (NLP) has become an essential technology in today's tech industry and is used in many real-world applications. One major use case is **chatbots and virtual assistants**, such as customer support bots and voice assistants like Siri and Google Assistant. These systems use NLP to understand user queries, provide relevant responses, and automate communication, improving customer experience and reducing operational costs. Another important use case is **sentiment analysis**, where NLP techniques are used

to analyze opinions and emotions expressed in text such as social media posts, product reviews, and customer feedback. Businesses use sentiment analysis to understand customer satisfaction, monitor brand reputation, and make data-driven decisions. A third major use case is **machine translation**, which enables automatic translation of text from one language to another. Services such as Google Translate use NLP to break language barriers and facilitate global communication. Besides these applications, NLP is also widely used in spam detection, text summarization, search engines, speech recognition, and recommendation systems. These use cases demonstrate how NLP helps computers understand human language, automate complex tasks, and improve interactions between humans and technology, making it one of the most important areas of artificial intelligence today.

Question 4: What is text normalization and why is it essential in text processing tasks?

Text normalization is the process of converting text into a standard and consistent format before it is analyzed by Natural Language Processing (NLP) systems. Raw text data often contains inconsistencies such as uppercase and lowercase letters, punctuation marks, special characters, abbreviations, spelling variations, and unnecessary spaces that can affect the performance of NLP models. Text normalization includes techniques such as converting all text to lowercase, removing punctuation and special symbols, expanding contractions (e.g., "don't" to "do not"), correcting spelling errors, removing extra whitespace, and converting words to their root forms through stemming or lemmatization. This process is essential because computers treat different forms of the same word as separate entities, which can lead to inaccurate analysis and increased complexity. By normalizing text, the data becomes cleaner, more structured, and easier for algorithms to process. It helps improve the accuracy of tasks such as sentiment analysis, text classification, machine translation, information retrieval, and chatbot systems. Text normalization also reduces vocabulary size, minimizes noise in the dataset, and ensures that similar words are represented consistently. Therefore, it is a crucial preprocessing step that enhances the efficiency and effectiveness of NLP applications.

Question 5: Compare and contrast stemming and lemmatization with suitable

Stemming and lemmatization are text preprocessing techniques used in Natural Language Processing (NLP) to reduce words to their base form, helping computers process text more efficiently. **Stemming** is a simple and fast method that removes prefixes or suffixes from words to obtain a root form, often without considering the word's actual meaning or grammatical structure. As a result, the output may not always be a valid dictionary word. For example, the words *playing*, *played*, and *player* may all be reduced to *play*, while *studies* may become *studi*.

Lemmatization is a more advanced technique that considers the context and grammatical meaning of a word before converting it to its base or dictionary form, known as a lemma. For example, *running* becomes *run*, *better* becomes *good*, and *studies* becomes *study*. While stemming is faster and requires fewer computational resources, it may produce less accurate results. Lemmatization is more accurate and meaningful but requires linguistic knowledge and more processing time. Therefore, stemming is often used when speed is

important, whereas lemmatization is preferred when accuracy and proper language understanding are required.

Question 6: Write a Python program that uses regular expressions (regex) to extract all email addresses from the following block of text:

“Hello team, please contact us at support@xyz.com for technical issues, or reach out to our HR at hr@xyz.com. You can also connect with John at john.doe@xyz.org and jenny via jenny_clarke126@mail.co.us. For partnership inquiries, email partners@xyz.biz.”

```
import re
```

```
text = """
```

```
Hello team, please contact us at support@xyz.com for technical issues,
```

```
or reach out to our HR at hr@xyz.com.
```

```
You can also connect with John at john.doe@xyz.org and jenny via
```

```
jenny_clarke126@mail.co.us.
```

```
For partnership inquiries, email partners@xyz.biz.
```

```
"""
```

```
# Regex pattern for email addresses
```

```
pattern = r'[a-zA-Z0-9._%+-.]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}'
```

```
# Extract all email addresses
```

```
emails = re.findall(pattern, text)
```

```
# Display extracted emails
```

```
print("Email Addresses Found:")
```

```
for email in emails:
```

```
    print(email)
```

Output -

Email Addresses Found:

support@xyz.com

hr@xyz.com

john.doe@xyz.org

jenny_clarke126@mail.co.us

partners@xyz.biz

Question 7: Given the sample paragraph below, perform string tokenization and frequency distribution using Python and NLTK:

“Natural Language Processing (NLP) is a fascinating field that combines linguistics, computer science, and artificial intelligence. It enables machines to understand, interpret, and generate human language. Applications of NLP include chatbots, sentiment analysis, and machine translation. As technology advances, the role of NLP in modern solutions is becoming increasingly critical.”

String Tokenization and Frequency Distribution using NLTK

```
import nltk
```

```
from nltk.tokenize import word_tokenize
```

```
from nltk.probability import FreqDist
```

```
# Download tokenizer (run once)
```

```
nltk.download('punkt_tab')
```

```
paragraph = """
```

```
Natural Language Processing (NLP) is a fascinating field that combines linguistics,  
computer science, and artificial intelligence. It enables machines to understand,  
interpret, and generate human language. Applications of NLP include chatbots,  
sentiment analysis, and machine translation. As technology advances, the role of NLP  
in modern solutions is becoming increasingly critical.
```

```
"""
```

```
# Tokenization
```

```
tokens = word_tokenize(paragraph)
```

```
print("Tokens:")
```

```
print(tokens)
```

```
# Frequency Distribution
```

```
freq_dist = FreqDist(tokens)
```

```
print("\nWord Frequency Distribution:")
```

```
for word, frequency in freq_dist.items():
```

```
    print(word, ":", frequency)
```

Output -

Tokens:

```
['Natural', 'Language', 'Processing', '(', 'NLP', ')', 'is', 'a', 'fascinating', 'field', 'that',  
'combines', 'linguistics', ',', 'computer', 'science', ',', 'and', 'artificial', 'intelligence', '.',  
'It', 'enables', 'machines', 'to', 'understand', ',', 'interpret', ',', 'and', 'generate', 'human',  
'language', '.', 'Applications', 'of', 'NLP', 'include', 'chatbots', ',', 'sentiment', 'analysis',  
, 'and', 'machine', 'translation', '.', 'As', 'technology', 'advances', ',', 'the', 'role', 'of',  
'NLP', 'in', 'modern', 'solutions', 'is', 'becoming', 'increasingly', 'critical', '.']
```

Word Frequency Distribution:

Natural : 1

Language : 1

Processing : 1

(: 1

NLP : 3

) : 1

is : 2

a : 1

fascinating : 1

field : 1

that : 1

combines : 1

linguistics : 1

, : 7

computer : 1

science : 1

and : 3

artificial : 1

intelligence : 1

. : 4

It : 1

enables : 1

machines : 1

to : 1

understand : 1

interpret : 1

generate : 1

human : 1

language : 1

Applications : 1

of : 2

include : 1

chatbots : 1

sentiment : 1

analysis : 1

machine : 1

translation : 1

As : 1

technology : 1

advances : 1

the : 1

role : 1

in : 1

modern : 1

solutions : 1

becoming : 1

increasingly : 1

critical : 1

Question 8: Create a custom annotator using spaCy or NLTK that identifies and labels proper nouns in a given text

Custom Proper Noun Annotator using spaCy

import spacy

Load English language model

nlp = spacy.load("en_core_web_sm")

text = """Natural Language Processing (NLP) is a fascinating field that combines linguistics,

computer science, and artificial intelligence. It enables machines to understand, interpret, and generate human language. Applications of NLP include chatbots, sentiment analysis, and machine translation. As technology advances, the role of NLP in modern solutions is becoming increasingly critical.

"""

```
# Process the text
```

```
doc = nlp(text)
```

```
print("Proper Nouns Identified:")
```

```
# Custom Annotator
```

```
for token in doc:
```

```
    if token.pos_ == "PROPN":
```

```
        print(f'{token.text} --> PROPER NOUN')
```

Output -

Proper Nouns Identified:

Natural --> PROPER NOUN

Language --> PROPER NOUN

Processing --> PROPER NOUN

NLP --> PROPER NOUN

NLP --> PROPER NOUN

NLP --> PROPER NOUN

Question 9: Using Genism, demonstrate how to train a simple Word2Vec model on the following dataset consisting of example sentences:

```
dataset = [ "Natural language processing enables computers to understand human language", "Word embeddings are a type of word representation that allows words with similar meaning to have similar representation", "Word2Vec is a popular word embedding
```

technique used in many NLP applications", "Text preprocessing is a critical step before training word embeddings", "Tokenization and normalization help clean raw text for modeling"]

Write code that tokenizes the dataset, preprocesses it, and trains a Word2Vec model using Gensim.

```
# Train a Word2Vec Model using Gensim
```

```
from gensim.models import Word2Vec
```

```
from nltk.tokenize import word_tokenize
```

```
import nltk
```

```
# Download tokenizer (run once)
```

```
nltk.download('punkt')
```

```
# Dataset
```

```
dataset = [
```

```
    "Natural language processing enables computers to understand human language",
```

```
    "Word embeddings are a type of word representation that allows words with similar meaning to have similar representation",
```

```
    "Word2Vec is a popular word embedding technique used in many NLP applications",
```

```
    "Text preprocessing is a critical step before training word embeddings",
```

```
    "Tokenization and normalization help clean raw text for modeling"
```

```
]
```

```
# Preprocessing and Tokenization
```

```
processed_data = []
```

```
for sentence in dataset:
```

```

sentence = sentence.lower() # Convert to lowercase

tokens = word_tokenize(sentence)

processed_data.append(tokens)


# Train Word2Vec Model

model = Word2Vec(

    sentences=processed_data,

    vector_size=100,

    window=5,

    min_count=1,

    workers=4

)

```

```

# Display vocabulary

print(model.wv.index_to_key)

```

Output -

```

['word', 'a', 'text', 'is', 'similar', 'representation', 'embeddings', 'to', 'language',
'modeling', 'for', 'raw', 'clean', 'help', 'normalization', 'and', 'tokenization', 'training',
'before', 'step', 'critical', 'preprocessing', 'applications', 'nlp', 'many', 'in', 'used',
'technique', 'embedding', 'popular', 'word2vec', 'have', 'meaning', 'with', 'words',
'allows', 'that', 'of', 'type', 'are', 'human', 'understand', 'computers', 'enables',
'processing', 'natural']

```

Question 10: Imagine you are a data scientist at a fintech startup. You've been tasked with analyzing customer feedback. Outline the steps you would take to clean, process, and extract useful insights using NLP techniques from thousands of customer reviews.

1. Data Cleaning

- Collect customer reviews from different sources.
- Remove duplicate reviews.
- Remove special characters, punctuation, URLs, and unnecessary spaces.

2. Text Preprocessing

- Convert text to lowercase.
- Perform tokenization (split text into words).
- Remove stop words.
- Apply stemming or lemmatization.

3. Sentiment Analysis

- Classify reviews as positive, negative, or neutral.
- Measure overall customer satisfaction.

4. Keyword and Frequency Analysis

- Identify frequently occurring words and phrases.
- Find common customer issues and concerns.

5. Topic Modeling

- Group reviews into topics such as payments, customer support, account issues, and app performance.
- Discover major discussion themes.

6. Named Entity Recognition (NER)

- Identify important entities such as product names, services, and company names mentioned in reviews.

7. Visualization and Insights

- Present results using charts and dashboards.
- Highlight customer satisfaction trends, common complaints, and areas for improvement.

8. Business Use

- Improve products and services.
- Enhance customer experience.
- Support data-driven business decisions.

